

Safe.Tech 2026: PQC Demo Playbook

This guide outlines the steps to perform a successful live demonstration of Post-Quantum Cryptography (PQC) using the STM32F769 Discovery Kit.

Goal of the Demo

Showcase the performance gap between classical RSA (3072/4096-bit) and modern PQC algorithms (Kyber/Dilithium) in a constrained embedded environment.

1. Preparation Phase (5 min before)

Hardware Setup

1. **Connect Board:** Use a Micro-USB or Mini-USB cable to connect the ST-LINK port of the **STM32F769I-DISCO** to your laptop.
2. **Verify LED:** Ensure the ST-LINK LED is solid (green/red) and the board is powered.

Software Environment

1. **Open PowerShell:** In the `pqc_demo` root directory.
 2. **Verify Tools:** Ensure you have Python and PowerShell execution permissions.
 3. **Find COM Port:** Check Device Manager to confirm the "STMicroelectronics STLink Virtual COM Port" is active (e.g., `COM5`).
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2. Scenario A: The "One-Click" Demo (Automated)

Best for a quick overview or during a presentation slot.

1. **Action:** Run the master session script:

```
powershell scripts/run_benchmarking_session.ps1
```

2. **What Happens:**

- The firmware is built and flashed instantly.
- The terminal displays live cycle counts for RSA-2048/3072/4096.
- **Highlight to Audience:** Notice the ~10-15s delay during RSA-4096 Key Generation.
- PQC algorithms (Kyber/Dilithium) will finish in milliseconds.

3. **Result:** The script automatically opens/updates `docs/FINAL_REPORT.md` and generates new plots in `docs/assets/`.
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3. Scenario B: Interactive Deep-Dive

Best for 1-on-1 booth conversations.

1. **Start Terminal:** Use the standard `run_demo.ps1` script:

```
powershell scripts/run_demo.ps1
```

2. **Explain the Options:**

- **Option 1 (RSA):** Run just RSA to show the "Legacy" struggle.
- **Option 2 (PQC):** Run the "Future" suite (Kyber/Dilithium). Point out the **pqm4 Assembly optimization** speedups.
- **Option 3 (All):** Full comparison.

3. **Live Interaction:** Let the visitor type '2' to see the PQC speed.

4. Key Talking Points ("The Wow Factor")

The RSA "Scaling Wall"

- **RSA-2048:** ~1.3s for Private Key Op. (Acceptable but slow).
- **RSA-4096:** ~12s for Private Key Op. (Too slow for real-time auth).
- **Point out:** Classical RSA scaling is **cubic**; as security needs go up, the performance drops off a cliff.

The PQC "Efficiency"

- **ML-DSA-44 (Dilithium):** Signing in **118ms**.
- **Comparison:** Dilithium is **>20x faster** than RSA-2048 and **>100x faster** than RSA-4096 on this hardware.
- **Assembly Gain:** Mention we use Cortex-M4/M7 assembly (`pqm4`) to gain another 2x speedup over standard C.

5. Handover

At the end of the demo, point the visitor to the `docs/FINAL_REPORT.md`:

"This report was just generated live from the board you see here. It includes the exact cycle counts and memory footprints for all candidates."

Emergency Reset: If the board hangs or UART stays silent, press the **Black Reset Button** on the board and restart the script.